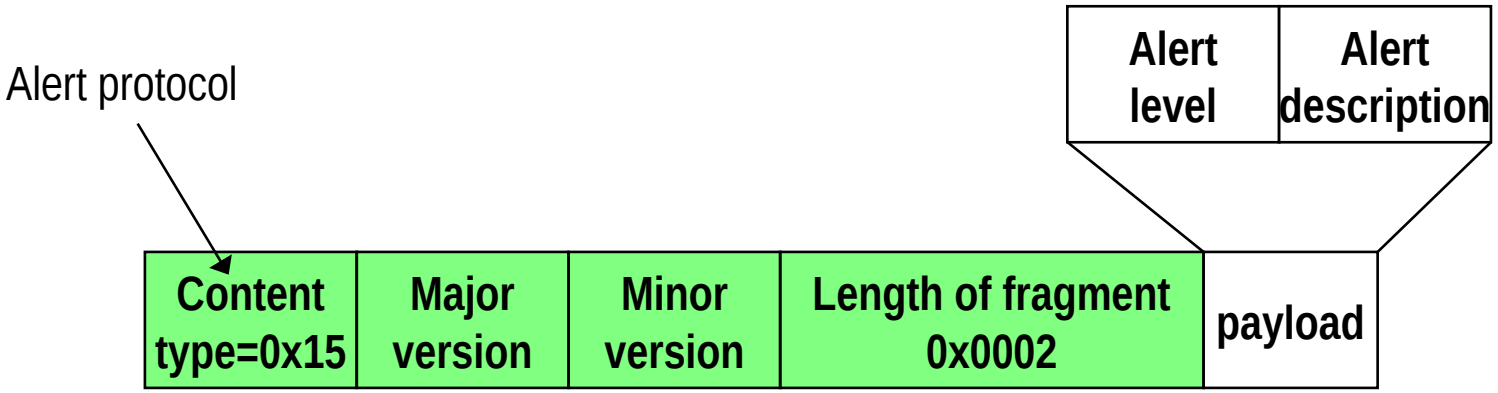
TLS connection management & application support

Alert Protocol

- **TLS defines special messages to convey "alert" information between the involved fields**
- **Alert Protocol messages encapsulated into TLS Records**
  - And accordingly encrypted/authenticated
- **Alert Protocol format (2 bytes):**
  - First byte: Alert Level
    - warning(1), fatal(02)
  - Second Byte: Alert Description
    - 23 possible alerts



TLS Record Header

Sample alerts

See RFC2246 for all alerts and detailed description

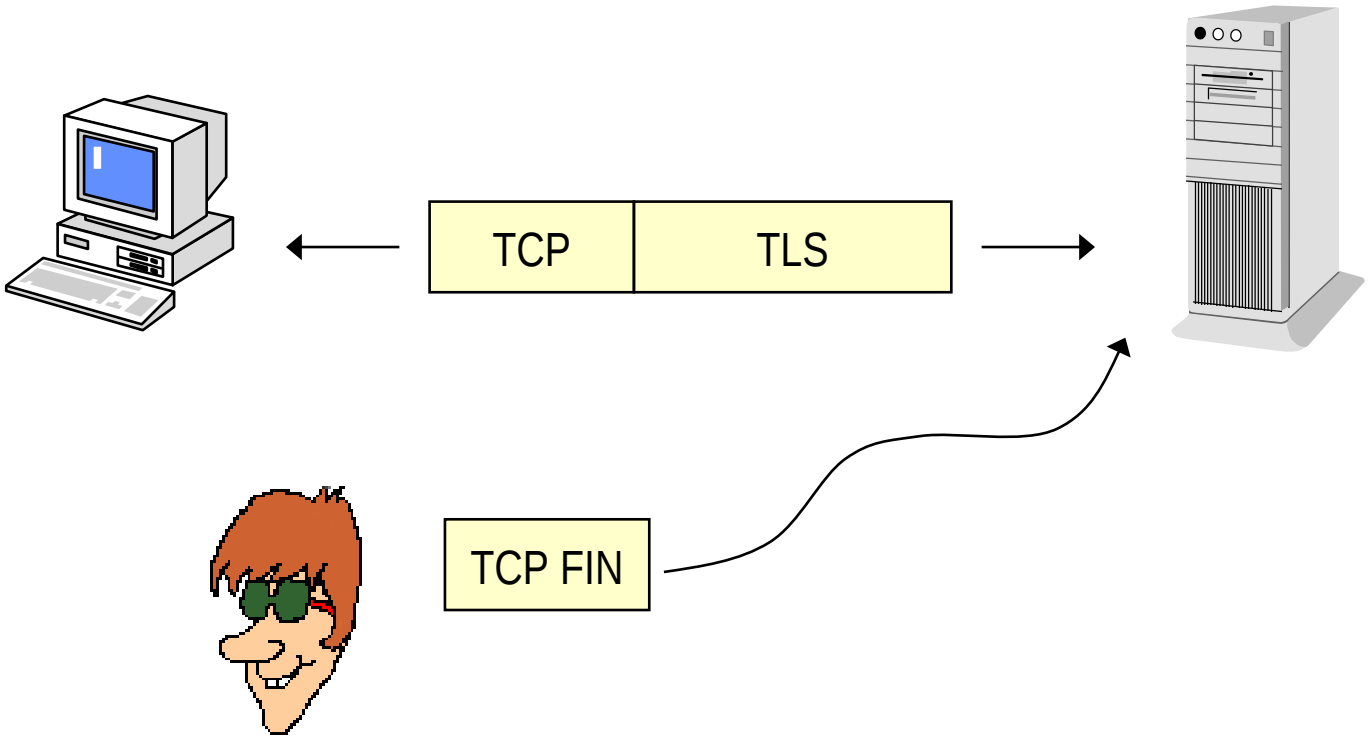
- **unexpected_message**
  - Inappropriate message received
    - Fatal - should never be observed in communication between proper implementations
- **bad_record_mac**
  - Record is received with an incorrect MAC
    - Fatal
- **record_overflow**
  - Record length exceeds 2^{14}+2048 bytes
    - Fatal
- **handshake_failure**
  - Unable to negotiate an acceptable set of security parameters given the options available
    - Fatal
- **bad_certificate, unsupported_certificate, certificate_revoked, certificate_expired, certificate_unknown**
  - Various problems with a certificate (corrupted, signatures did not verify, unsupported, revoked, expired, other unspecified issues which render it unacceptable
    - Warning or Fatal, depends on the implementation

If fatal, terminate and do not allow resumption with same security parameters (clear all!)

Remark: TLS do NOT protect TCP

- **Open to DoS attacks**
  - Anyone can kill a TCP connection
    - Just spoof a RST…
- **How to protect non SSL/TLS-aware applications?**
  - stunnel = TCP over TLS over TCP (!!!)
    - www.stunnel.org
    - Use as LAST resort…
  - TCP over TCP tunnels: performance impairment!!
    - Conflicting congestion control loops
    - Use DTLS tunnels, instead

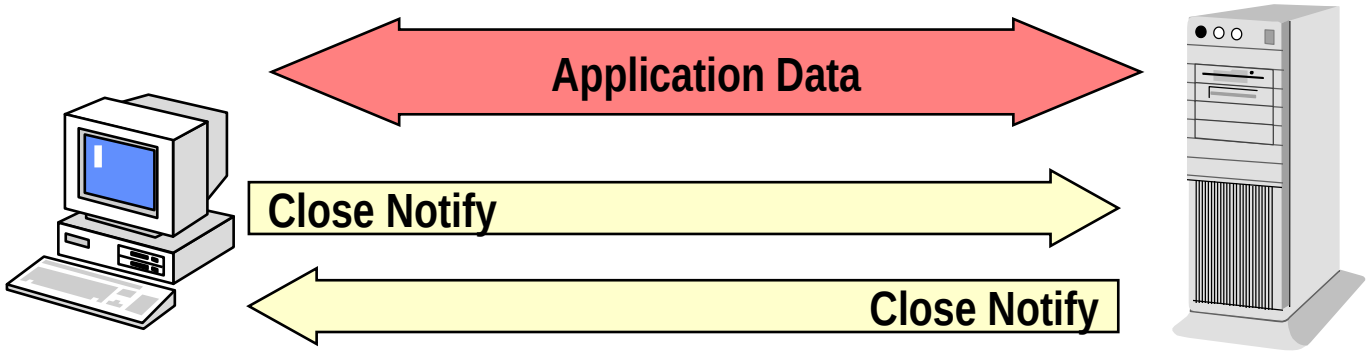
Truncation attack



An attacker may end connection at any time by sending a TCP FIN
Part of the intended information exchange is lost

How can Server and Client distinguish this from a transaction that “normally” completes?

Solution: Close Notify

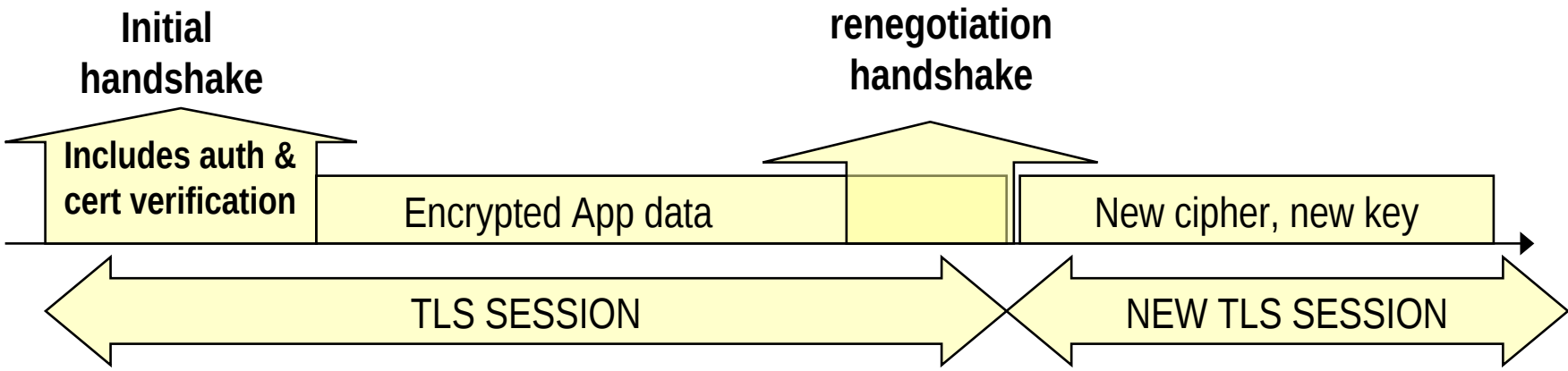


- **Issued by any party**
  - Close Notify = Alert (warning level)
  - Half-close semantics
- **Informs that no more data will be transmitted**
- **A connection that ends abruptly without a Close Notify may be a truncation attack**

A remark: note the weakness of TLS against TCP DOS attacks in general!

Renegotiation

- **Renegotiation (re-handshake)**
  - Not "limited to" rekeying; more than session resumption
- **New full handshake (optionally) permitted.**
  - Encrypted via previously established ciphersuite
- **Purpose:**
  - Increase the security level (e.g. move to higher security cipher)
  - Authenticate Client
  - Anything else you may need
- **Renegotiation generates a NEW session ID**



Important: besides the fact that it is encrypted, renegotiation handshake otherwise NOT distinguishable from initial handshake (session = 0) – can this fact be (ab)used?

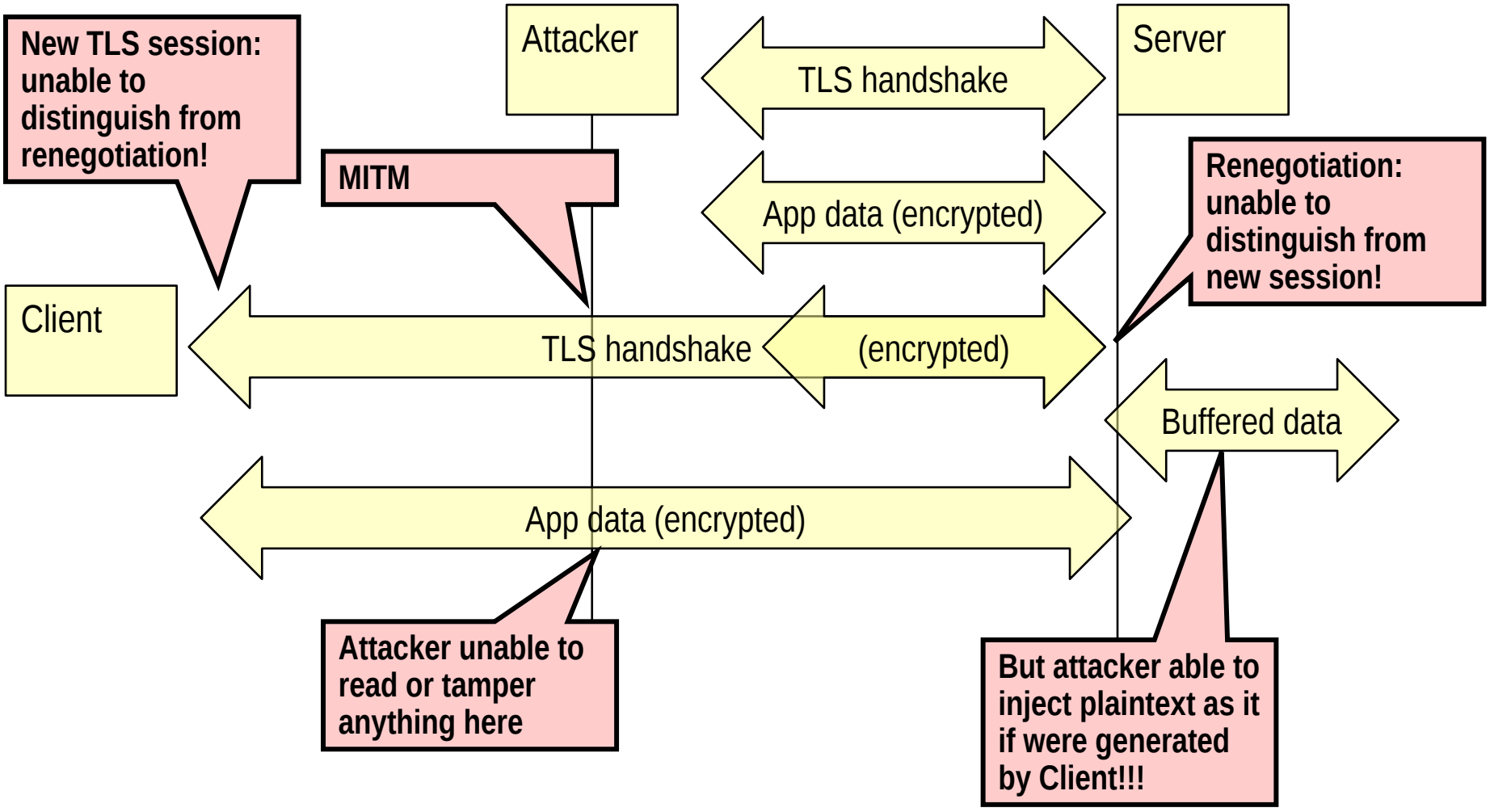
What about application data?

- **Renegotiation, when required, takes precedence over application data**
  - If applicable, server will buffer application data until renegotiation completed, then proceed on NEW TLS session
- **… renegotiation can be in the middle of an application layer transaction…**
  - Can we exploit this???

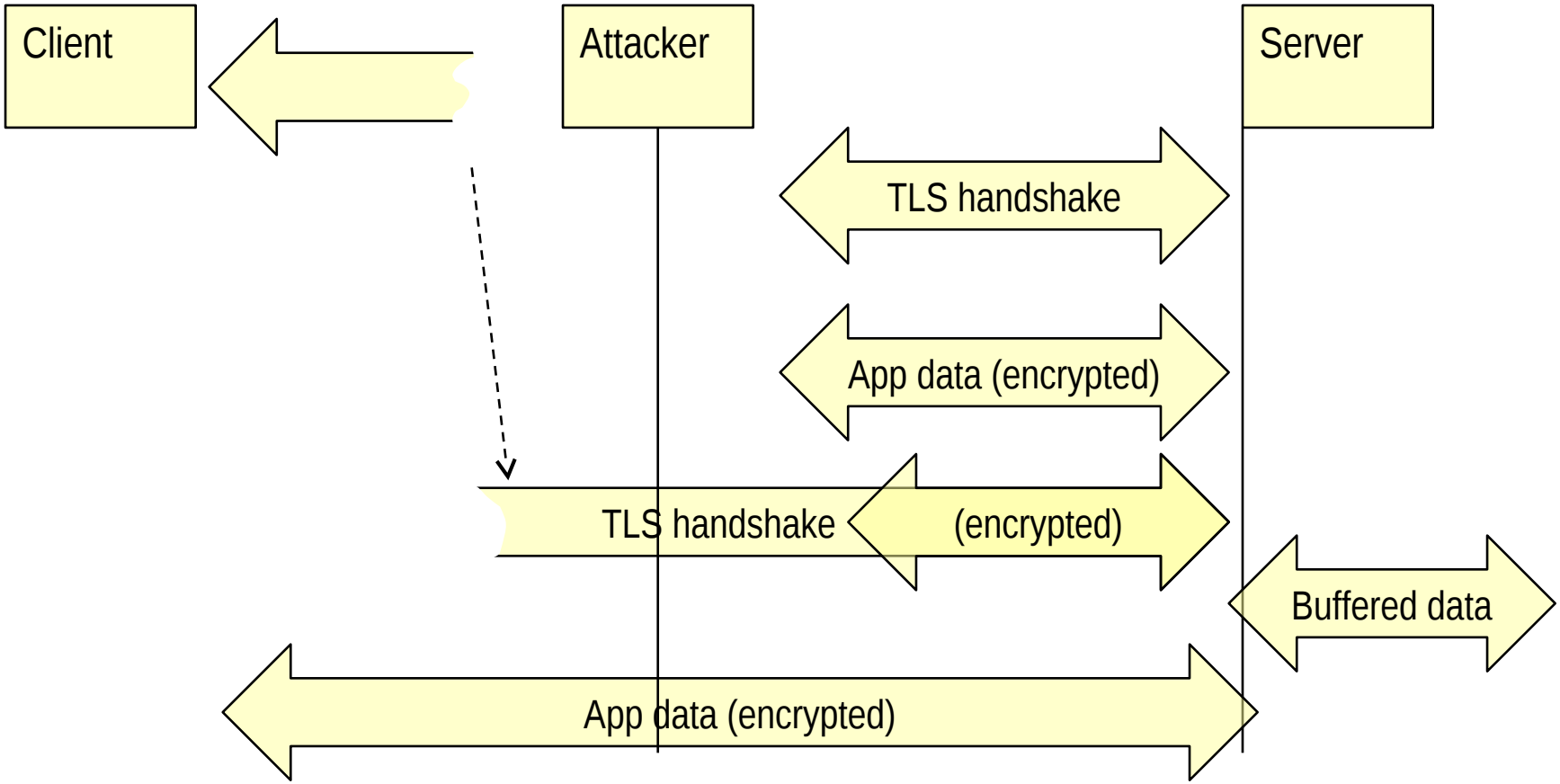
Renegotiation attack
(plaintext injection attack)

- **Discovered by Marsh Ray (PhoneFactor inc.)**
  - Aug-sep 2009; made public on begin of nov 2009
  - Limited effectiveness… but
- **Turned into practical credential stealing attack against twitter users by Anil Kurmus (ETH student)**
  - Mid november 2009
  - Twitter fixed it immediately (by disallowing renegotiation)
  - Ingenious application of such attack may apply to many other cases/scenarios
- **Came as a shock for TLS standard bodies**
  - Take-home golden rule: in security KEEP things SIMPLE!!

How it works


New TLS session: unable to distinguish from renegotiation!
Attacker
TLS handshake
Server
MITM
App data (encrypted)
Renegotiation: unable to distinguish from new session!
Client
TLS handshake
(encrypted)
Buffered data
App data (encrypted)
Attacker unable to read or tamper anything here
But attacker able to inject plaintext as it if were generated by Client!!!

How it works / bis



Non trivial details (omitted here): i) how to properly buffer data? Ii) how to precisely trigger renegotiation? Both can be sorted out (indeed, attack was experimentally proved)

What can we do with prefix data injection?

- **Depends on application, of course! Example:**

- **Attacker:**
  - GET /pizza?toppings=pepperoni;address=attacker_address HTTP/1.1
  - X-Ignore-This: (no carriage return)

- **Victim:**
  - GET /pizza?toppings=sausage;address=victim_address HTTP/1.1
  - Cookie: victim_cookie

- **Result: glued requests:**
  - GET /pizza?toppings=pepperoni;address=attacker_address HTTP/1.1
  - X-Ignore-This: GET /pizza?toppings=sausage;address=victim_address HTTP/1.1
  - Cookie: victim_cookie

- **Server uses victim's account to send a pizza to attacker!**

Nicer example

(getting closer to actual Kurmus' twitter attack)

- **Attacker:**
  - POST /forum/send.php HTTP/1.0
  - Message = (no carriage return)
- **Victim:**
  - GET / HTTP/1.1 […]
- **Result: glued requests:**
  - POST /forum/send.php HTTP/1.0
  - Message = GET / HTTP/1.1 […]
- **Server discloses information generated by client upon request**
- **Anil Kurmus attack: conceptually similar**
  - Technicalities related to twitter API
  - Result (real world!): server posts a tweet (!) including the client credentials (base64 coded user/pass)!!!

Preventing TLS reneg attack

- **Immediate solution: disable it**

- **Standardization patch: renegotiation extension**
  - (urgently) standardized in RFC 5746
    - "TLS renegotiation Extension", February 2010
  - First message: Extension empty
  - Second message: Extension carries previous handshake verify (finished msg)
    - Now it is trivial to detect attack

- **TLSv1.3: forbids renegotiation!**
  - «keep it simple» rule

And what about DTLS?

DTLS vs TLS at a glance

- **RFC 4347 – Datagram TLS – April 2006**
  - TLS over UDP
- **DTLS design goal:**
  - Be as most as possible similar to TLS!
- **DTLS vs TLS**
  - TLS assumes orderly delivery
    - » DTLS: Sequence number explicitly added in record header
  - TLS assumes reliable delivery
    - » Timeouts added to manage datagram loss
  - TLS may generate large fragments up to 16384B
    - » DTLS includes fragmentation capabilities to fit into single UDP datagram, and recommends Path MTU discovery
  - TLS assumes connection oriented protocol
    - » DTLS connection = (TLS handshake – TLS closure Alert)